Luis Ramirez

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Plainfield, NJ 07060

Education

New Jersey Institute of Technology (NJIT) — Newark, NJ
Bachelor of Science in Mechanical Engineering — *GPA 3.24 — Expected Dec 2027*

- Psychology Minor and BS/MS Dual-Degree Program (Master of Science) — In Progress
- Active member, Educational Opportunity Program (EOP) — Jun 2023 – Present

Skills & Coursework

Technical & Professional Skills: 3D CAD & drawings, prototyping & design iteration, GD&T & inspection, root cause analysis & troubleshooting, data analysis, technical reports, quality & safety procedures, bilingual Spanish
Software: SolidWorks & Creo Parametric (3D modeling, drawings, BOMs), SolidWorks Simulation, MATLAB, Microsoft Excel & Office
Relevant Coursework: Machine Design, Experimental Methods, Fluid Mechanics, Heat Transfer, Stress Analysis

Projects

Machine Vise CAD Modeling & Drawing Package | *SolidWorks* — Sep 2026 – Nov 2026

- Worked on a 5-person team that modeled an 8-component bench vise as fully constrained parametric parts in SolidWorks and mated them into a working assembly that opens and closes through its full travel without rework.
- Helped produce a 12-sheet ANSI A drawing set with section and detail views (knurl, ACME 5/8"-8 thread) and an exploded assembly with a bill of materials, and checked teammates' sheets before submission.

Gyroscopic Cup Holder | *SolidWorks, 3D Printing* — Sep 2025 – Dec 2025

- Designed a passive, self-leveling gimbal cup holder in SolidWorks for cyclists and wheelchair users (fits 64–100 mm cups, under 1 kg), benchmarking 3 products and 3D printing a PLA prototype.
- Hand-tilt tested the prototype, found the fits too tight, and revised the pin-hole diameter from 0.175 to 0.2 in and eased the rib fit by 5 degrees until it assembled.

Experience

Automotive Parts Specialist — Jul 2025 – Present
AutoZone Auto Parts — *Newark, NJ*

- Perform hands-on installation, testing, and troubleshooting of automotive components including 50+ batteries, 100+ lighting components, wipers, fuses, spark plugs, fluids, and air/cabin filters using battery testers, OBD-II scanners, torque tools, and other automotive equipment, while explaining findings and corrective actions.
- Apply vehicle-specific specifications including year, make, model, engine, trim, and part numbers to determine component fitment and diagnose issues; inspect for wear, leakage, overheating, and improper installation.

Field Engineering Technician — Aug 2021 – Apr 2023
Pittenger Paving & Excavating, Inc. — *Dunellen, NJ*

- Measured and verified 20+ construction sites using tape measures and laser-level equipment to confirm layout accuracy and structural tolerances, supporting proper installation of infrastructure components and execution.
- Operated heavy machinery including skid-steers, rollers, and tampers while assisting with paving and excavation projects, maintaining safety procedures and supporting efficient material placement in demanding environments.

Leadership & Activities

EOP STEM Tutor — Jul 2024 – May 2025
New Jersey Institute of Technology (NJIT) — *Newark, NJ*

- Delivered 200+ hours of academic tutoring to first-year students in calculus, physics, and engineering, breaking down complex problem sets and exam preparation step by step.
- Facilitated 2-hour group study sessions and workshops 2–3 times per week in-person and via Discord, adapting problem-solving methods to individual student needs.

Alpha Phi Omega (APO) — Professional Service Fraternity — Sep 2025 – Present
Active Brother (NJIT) — *Newark, NJ*

- Completed 60+ hours of hands-on community service, including local animal shelter support in Montclair, community blood drives, and clothing donations.
- Collaborated with active brothers across 30+ chapter meetings to plan events, manage volunteer shifts, and support local Newark and regional outreach initiatives.